

What the symbols actually mean

The three thermodynamics formulas, taken apart one squiggle at a time, then used. Every value checked before it was printed.

1 Every symbol, before any formula

These formulas look frightening because they are written in shorthand. Each squiggle is an instruction. Once each one is read out loud in English, the formula is just a shopping list.

Symbol	Said out loud	What it actually tells you to do
Δ	delta	Means change in . Not a number and not a thing you multiply by — it just says the next letter is a difference : end minus start.
H	enthalpy	The heat energy stored in the chemicals. On its own it is never used; it always turns up as ΔH .
ΔH	delta H	The heat given out or taken in by the reaction. Negative = heat comes OUT (hot). Positive = heat goes IN (cold).
Σ	sigma	Greek capital S, and S is for sum . It means add up all of them . That is the whole meaning. It never means multiply.
n	n	The big number in front of a substance in the balanced equation. In $2\text{H}_2\text{O}$ the n is 2 . If nothing is written, $n = 1$.
$\Delta_f H$	delta f H	Enthalpy of formation . It is a number you look up in the table the question gives you. You never calculate it. The little f is just a label meaning "formation".
(prod)	products	Do this bit for everything on the RIGHT of the arrow.
(react)	reactants	Do this bit for everything on the LEFT of the arrow.
S°	S standard	Entropy — how disordered the substance is. Also just a number you look up. Gases have big values, solids small ones.
ΔS	delta S	The change in disorder . Positive means the reaction made things messier, which usually means it made a gas.
ΔG	delta G	Free energy . The one that answers "will this reaction happen by itself?" Negative = yes.
T	T	Temperature, and it must be in kelvin . $K = ^\circ C + 273$, every single time.

2 The enthalpy formula, taken apart

$$\Delta H = \Sigma n \Delta_f H (\text{products}) - \Sigma n \Delta_f H (\text{reactants})$$

Read left to right, it says exactly this:

Symbol	Said out loud	What it actually tells you to do
$\Delta H =$	the answer	the heat change you are being asked for

Σ	add up	go through every substance on the right of the arrow, one at a time
n	times its big number	for each one, take its coefficient from the balanced equation
$\Delta_f H$	times its table value	multiply that by the number the question gives you for that substance
—	then subtract	now do the whole thing again for the left side, and take it away

THE ONE SENTENCE VERSION

For every substance, multiply **the big number in front** by **the number from the table**. Add up the **right-hand side**. Add up the **left-hand side**. Subtract.

WHY A TABLE BEATS THE FORMULA

The Σ is only telling you to add up a column. So make the column. One row per substance, and the last column is the only thing you add. Every worked example below is done this way.

3 Worked example — enthalpy

Question. Calculate the enthalpy change of reaction for the combustion of methane.



Given: $\Delta_f H(\text{CH}_4) = -74.8$, $\Delta_f H(\text{CO}_2) = -393.51$, $\Delta_f H(\text{H}_2\text{O}) = -241.82$ kJ/mol.

Step 1. Split the equation down the middle. Anything on the **left** is a reactant; anything on the **right** is a product. Write every substance into a table, one per row.

Step 2. Fill in n , which is just the big number in front. CH_4 has nothing written, so $n = 1$. O_2 has a 2. And so on.

Step 3. Fill in $\Delta_f H$ by copying the number the question gave for that substance.

WHY O_2 IS NOT IN THE LIST

Oxygen is an **element on its own**, and every element in its normal state has $\Delta_f H = 0$. It is missing from the given values because it does not need one. Write 0 in its row and carry on.

Step 4. Multiply across each row, then add up the last column for each side.

Substance	Side	n	$\Delta_f H$	$n \times \Delta_f H$
CO_2	product	1	-393.51	-393.51
H_2O	product	2	-241.82	-483.64
CH_4	reactant	1	-74.8	-74.8
O_2	reactant	2	0	0
SUM of the products				-877.15
SUM of the reactants				-74.8

Step 5. Subtract: products minus reactants.

$$\Delta H = (-877.15) - (-74.8)$$

$$\Delta H = -877.15 + 74.8$$

THE STEP THAT CATCHES EVERYONE

Subtracting a negative number **adds** it. The minus in the formula and the minus already on -74.8 cancel each other out. If you type $-877.15 - 74.8$ into the calculator you get -951.95 , which is wrong. Put the reactant total in **brackets** and let the calculator sort out the signs.

$$\Delta H = -802.35 \text{ kJ}$$

DOES THE ANSWER MAKE SENSE?

It is **negative**, so heat comes out. Burning methane is a gas flame, so of course it does. A positive answer here would have meant a mistake, and that sanity check is free.

4 Worked example — entropy

Exactly the same shape of table. The looked-up value is called S° instead of $\Delta_f H$.

THE ONE DIFFERENCE THAT TRIPS PEOPLE

For entropy, elements are **NOT** zero. Every substance has a real S° value, including N_2 and H_2 . That rule about elements being zero belongs to enthalpy only.

$$\Delta S = \sum n S^\circ (\text{products}) - \sum n S^\circ (\text{reactants})$$

Question. Find ΔS for the Haber process.



Given: $S^\circ(N_2) = 191.5$, $S^\circ(H_2) = 131.0$, $S^\circ(NH_3) = 193.0 \text{ J mol}^{-1}\text{K}^{-1}$.

Substance	Side	n	S°	$n \times S^\circ$
NH_3	product	2	193.0	386.0
N_2	reactant	1	191.5	191.5
H_2	reactant	3	131.0	393.0
SUM of the products				386.0
SUM of the reactants				584.5

$$\Delta S = 386.0 - 584.5$$

$$\Delta S = -198.50 \text{ J mol}^{-1}\text{K}^{-1}$$

DOES THE ANSWER MAKE SENSE?

Four molecules of gas went in ($1 + 3$) and only two came out. Fewer gas molecules means **more order**, so ΔS should be negative. It is.

5 Worked example — free energy

This one needs no table at all. There are only three numbers, and they go straight in.

$$\Delta G = \Delta H - T \Delta S$$

Question. For the same Haber reaction, $\Delta H = -92.2 \text{ kJ}$ and $\Delta S = -198.5 \text{ J mol}^{-1}\text{K}^{-1}$. Is it spontaneous at 25°C ?

Step 1 — temperature to kelvin.

$$T = 25 + 273 = 298 \text{ K}$$

Step 2 — make the units match. ΔH is in **kJ** and ΔS is in **J**. They cannot be combined until they agree, so divide the entropy by 1000.

$$\Delta S = -198.5 \div 1000 = -0.1985 \text{ kJ mol}^{-1}\text{K}^{-1}$$

THIS IS THE MOST COMMONLY DROPPED MARK IN THE TOPIC

Entropy tables are always printed in **J**. Enthalpy is always in **kJ**. Miss the division and the answer comes out about a thousand times too big, and the verdict flips to the wrong one.

Step 3 — substitute. Put each number where its letter is.

$$\Delta G = (-92.2) - (298)(-0.1985)$$

Step 4 — the multiplication first.

$$(298)(-0.1985) = -59.153$$

$$\Delta G = -92.2 - (-59.153)$$

$$\Delta G = -92.2 + 59.153$$

THE DOUBLE NEGATIVE AGAIN

A minus sign in front of a negative number turns it into a plus. It happens here for the same reason it happened in the enthalpy example. Whenever ΔS is negative, that middle term ends up **adding**.

$\Delta G = -33.05 \text{ kJ/mol} \rightarrow$ negative, so the reaction IS spontaneous at 298 K

WHAT THE SIGN IS TELLING YOU

ΔG below zero means the reaction goes by itself. Above zero means it will not. Exactly zero means it is balanced at equilibrium.

The follow-up they usually ask next

"At what temperature does it stop being spontaneous?" means set ΔG to zero and solve for T .

$$0 = \Delta H - T\Delta S \quad \text{so} \quad T = \Delta H \div \Delta S$$

$$T = (-92.2) \div (-0.1985) = 464.5 \text{ K}$$

WHICH SIDE OF THAT TEMPERATURE?

Both ΔH and ΔS are negative here. The rule: when the two signs **match**, the reaction is spontaneous on one side of the crossover only. Both negative means spontaneous **below** 464.5 K — and 298 K is below it, which agrees with the answer above.

6 The whole thing on one card

Symbol	Said out loud	What it actually tells you to do
ΔH	heat	Table of $\Delta_f H$. Products minus reactants . Elements are zero . Answer in kJ.

ΔS	disorder	Table of S° . Products minus reactants . Elements are NOT zero . Answer in J.
ΔG	will it go	No table. $\Delta G = \Delta H - T\Delta S$, T in kelvin, entropy converted to kJ first.

THE THREE QUESTIONS TO ASK YOURSELF EVERY TIME

1. Is my temperature in kelvin? 2. Are enthalpy and entropy both in kJ? 3. Did I subtract a negative and remember it becomes a plus?

AND THE SANITY CHECK

Burning things gives out heat, so combustion ΔH is negative. Making gas raises disorder, so ΔS is positive. If your answer disagrees with common sense, it is usually a sign, not the chemistry.